System Name: Canteen Pre Order and Queue Management System
Group Members: Borreta, Joshua; Bola, Martin Lawrence; Juatchon, Joaquin Phillip; Marcos, Karl Raphael; Pacheco Jr, Sancho; Vera Cruz, Euan Joseph
Repository Link:
https://github.com/EclipseChan/bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System

Member Roles:

1. Euan Vera Cruz - Project Manager, Developer
2. Karl Marcos - Lead Developer
3. Joaquin Phillip Juatchon - Quality Assurance
4. Joshua Borretta - Developer, and Quality Assurance
5. Martin Lawrence Bola - Technical Documentation
6. Sancho Pacheco Jr - Technical Documentation

System Overview
As per its title, this application is a Canteen Pre Order and Queue Management System, where Students and Vendors can register their accounts and log in to them. Students can order from listed Vendors, see the pricing and status of their orders. Vendors can set up their menu items, change the status of a Student's order and see their sales for the day.

Milestones / Day-to-Day Progress:

October 31	Creation of initial backend uploaded to Github Repository. Had incomplete functionalities, but was still able to produce outputs. Files were made to allow roles to be made, though lacked most functions it needed. Created a file for groupmates to allow manual setup of the database since we lack automated creation.
November 1	No uploads within the Repository today, though created a test application that contained most backend functionality. Created a separate, private repository for testing purposes.
November 2	Using current code and the test application, we made the backend fully functional as far as we can tell, containing most, if not all of what was within our

	proposal. Originally had none of the scripts, docs, and other assets needed, though was added later in the day's development. Github upload was done a few times to properly update files needed for it to work, along with the ReadME for directions on setup.
November 3	Creation of Frontend Interface, it uses React for access, testing and it works for most of its functionalities, such as logging into vendor and student accounts.
November 4	Finalization of functional application, and creation of branches set for the following activities 4 to 6. Subsequent Midterms branch set to be made soon.
November 5	Application made compliant with the criteria of Activity 4, 5 and 6. Added test results from Groups 2, 3, 4, 6, 7
November 6	Continued testing for remaining groups. Revised tests of some groups due to testing errors.

Contributors and Contributions:

1. Euan Vera Cruz - headed the main development, main contributor in Github.
2. Karl Marcos - provided scripts, docs, and database markdown files along with the initial proposal idea from Activity 2. Made application compliant with activities 4-6.
3. Joaquin Juatchon - tested and confirmed application functions on a different computer that has not seen the application.
4. Joshua Borreta - tested and confirmed tests with other groups' applications. Notably Group 5.
5. Martin Lawrence Bola - confirmed functionality of main repo, and tested group 10's application, though his initial findings were passed on over and continued by other members.
6. Sancho Pacheco Jr. - tested Group 6's application and confirmed functionality, though member Vera Cruz headed the tests.

Identified Issues and Resolutions:

1. Java 25 does not work - the pom xml file explicitly wants Java 21, though it can be updated, it has not been tested. A solution is given in the ReadME.
2. NPM policy - if npm install or npm run dev doesn't work, use npm.cmd with install and run dev.
3. PSQL unrecognized - add its BIN folder to PATH, and test with a new terminal.
4. Missing ENV file - run this env copy command from the ReadME, modify the lines that have "change-me."

Although these are the main issues identified during development, more issues are shown within the ReadME during the testing phase.

Application Requirements:

1. PostgreSQL 18 - for the database.
2. Java 21 - can be updated to 25, though has not been tested, so 21 is the sweet spot.
3. Maven - important to help run backend with Java 21.
4. Node JS - important for the frontend.

Changes from Proposal:

1. Lack of editing of order from Student - having a student edit an order mid preparation is conflicting, since the vendor is likely already preparing for it, and several factors hang off an order within it.
2. Database name - blueprint originally suggested grpxx_canteendb, implementation makes canteen_db, it's a minor change. But canteen_db makes it more simple.
3. Added Vendor credentials - although the proposal had vendors not have them, they were added since they are important for Vendors to manage their menus and orders.
4. Student.role was added - the proposal named administrators as users, but they lacked a table at the ERD. An admin is a student row with the role ADMIN, avoiding creating an extra table for one extra attribute.

AI Usage Declaration

1. Claude - was used to assist in creating the application, and tests. Developers reviewed code and adjusted based on proposal demands. It assisted in creating the markdown files for the ReadME, creation of tests that were reviewed by developers.
2. ChatGPT - was used for fixing errors involved with the process of starting the application.

3. Copilot (VSCode Agent) - used for automation of tests for activities. Although scripts used did most of the work, developers manually adjusted SQL parameters, ran commands through powershell so testing could be completed.

All AI generated and assisted codes and scripts were verified, adjusted, and tested by developers, ensuring tests would go properly with manual help.

Testing of Other Groups Tracker:

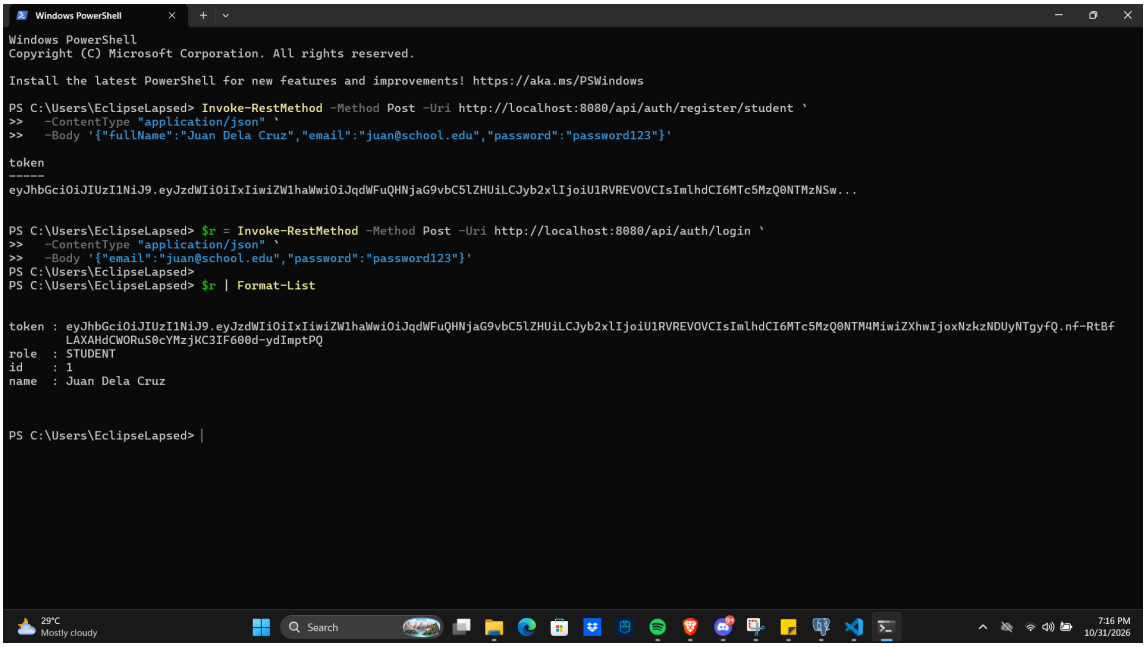
Group Name	**Application Name**	**Activity Status**
Group 2	Park It: Smart Parking System	Activity 3: Completed Activity 4: Completed Activity 5: Completed Activity 6: Completed
Group 3	OJT Record Management System	Activity 3: Completed Activity 4: Completed Activity 5: Completed Activity 6: Completed
Group 4	BrewPOS	Activity 3: Completed Activity 4: Completed Activity 5: Completed Activity 6: Completed
Group 5	PUP-SPOT	Activity 3: Completed Activity 4: Completed Activity 5: Completed Activity 6: Completed
Group 6	HabalGO	Activity 3: Completed Activity 4: Completed Activity 5: Completed Activity 6: Completed
Group 7	CAMPUSAFE	Activity 3: Completed Activity 4: Completed Activity 5: Completed Activity 6: Completed
Group 10	MESA	Activity 3: Completed Activity 4: Completed Activity 5: Completed Activity 6: Completed

Group 11	BoardTrack	Activity 3: Completed Activity 4: Completed Activity 5: Completed Activity 6: Completed
Group 12	iBarangay	Unable to test, lack of application link in sheet.

Will be updated if more groups are updated in the main tracker sheet, or if tests are complete.

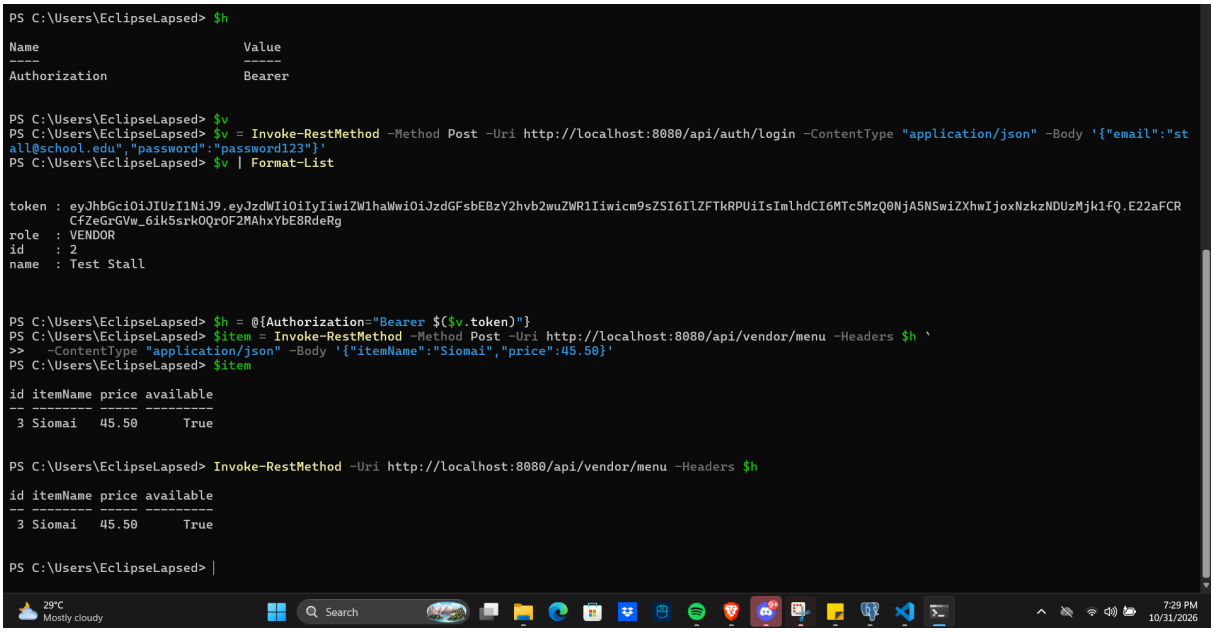
Picture Evidence:

1. October 31st:

a. 

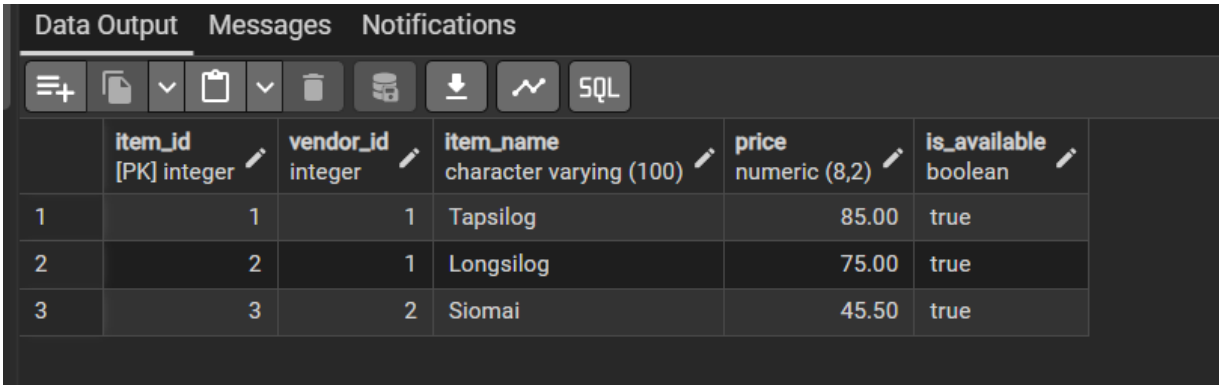
(Powershell demonstration showing creation of STUDENT role account)

b.



(Powershell test conducted showing VENDOR and MENU showcase, and showing a slight change in menu item's availability)

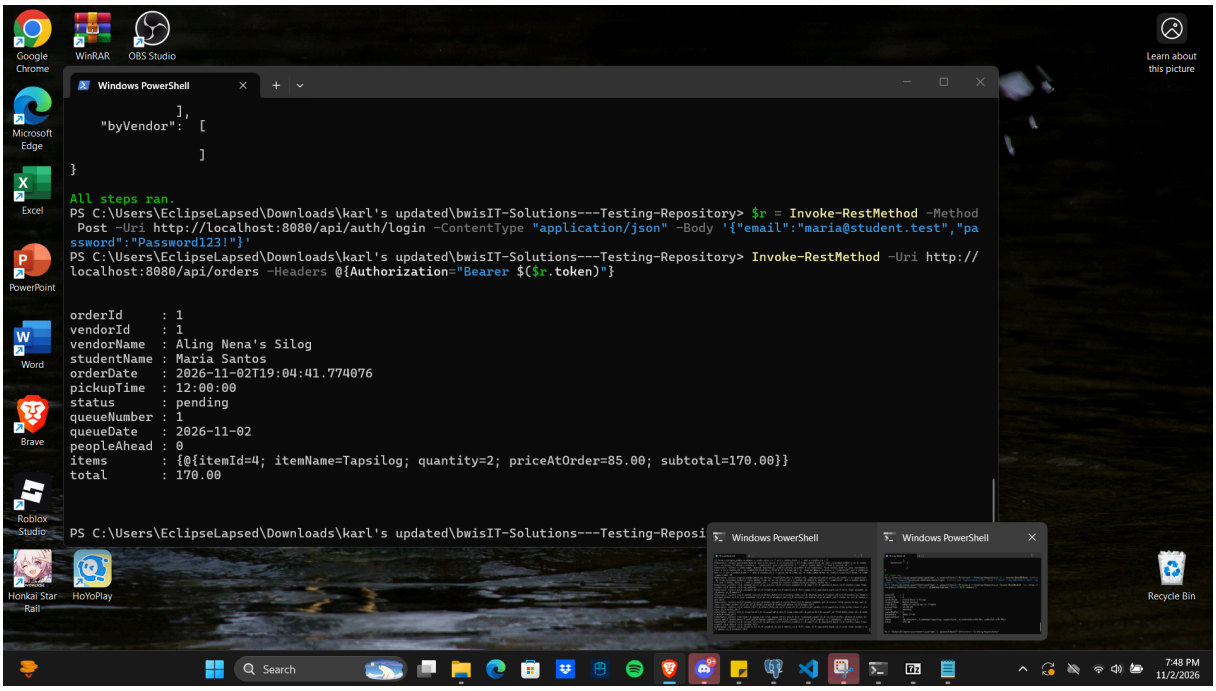
c.



	item_id [PK] integer	vendor_id integer	item_name character varying (100)	price numeric (8,2)	is_available boolean
1	1	1	Tapsilog	85.00	true
2	2	1	Longsilog	75.00	true
3	3	2	Siomai	45.50	true

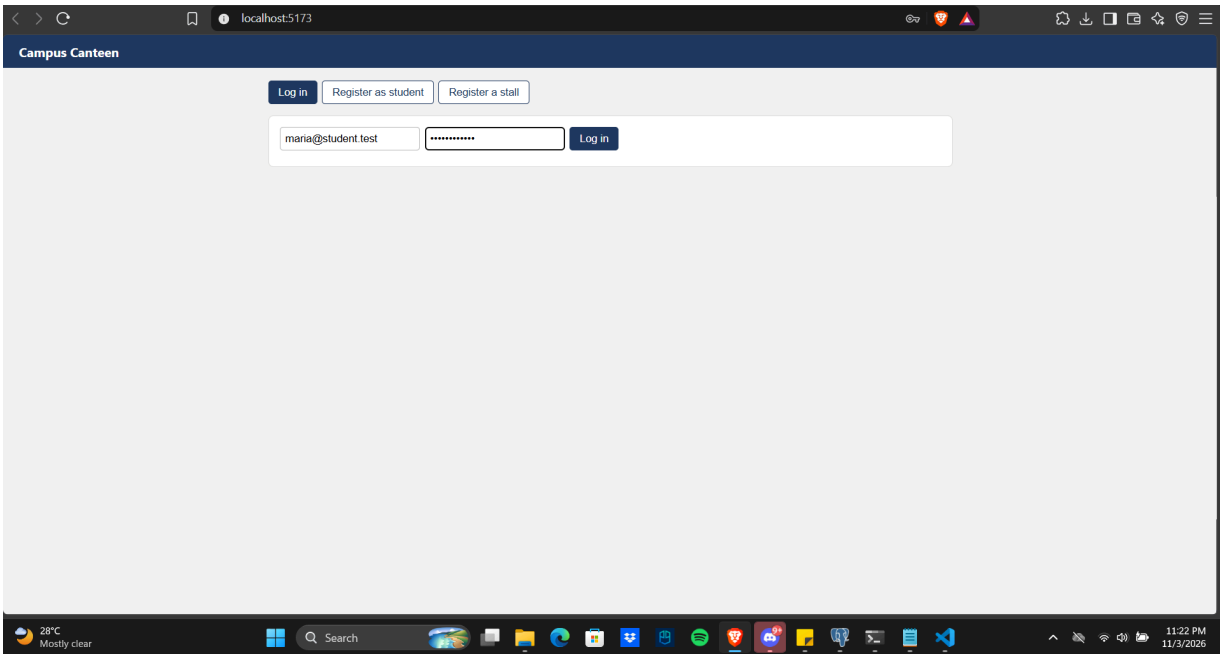
(From an earlier image, showcase of table within PostgreSQL)

2. November 2nd

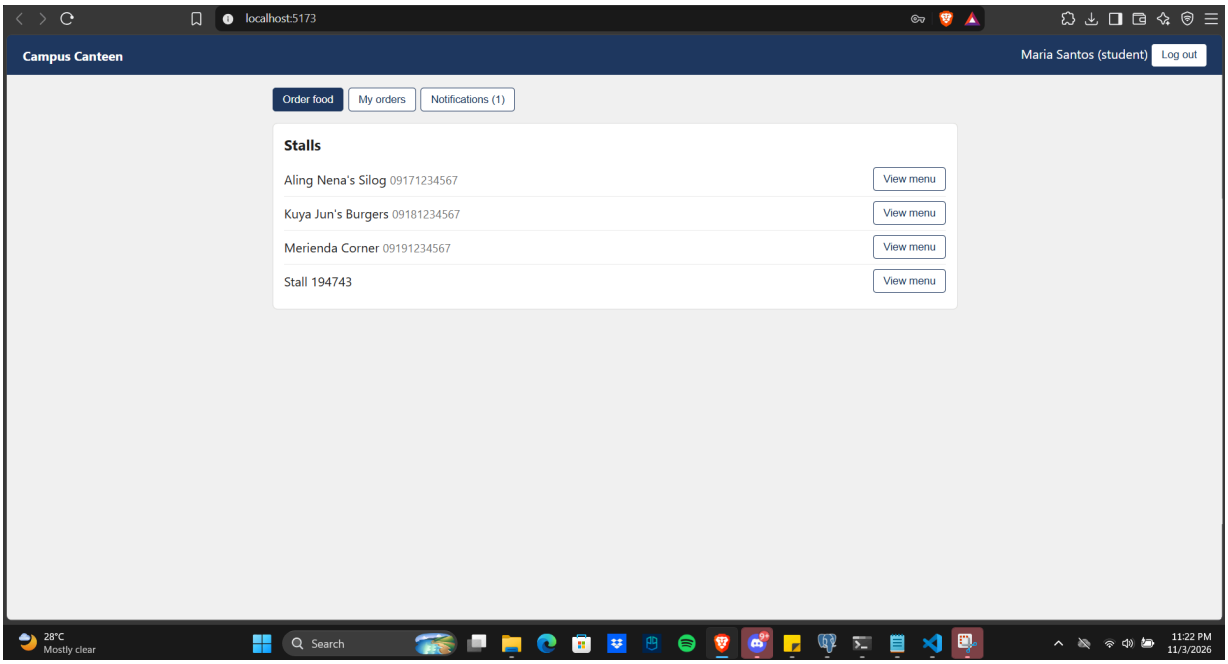
a. 

(Smoke test conducted on the host machine using test-flow.ps1 file within the repository. Two PowerShell windows are open, one with the application start, and the second being the output window)

3. November 3

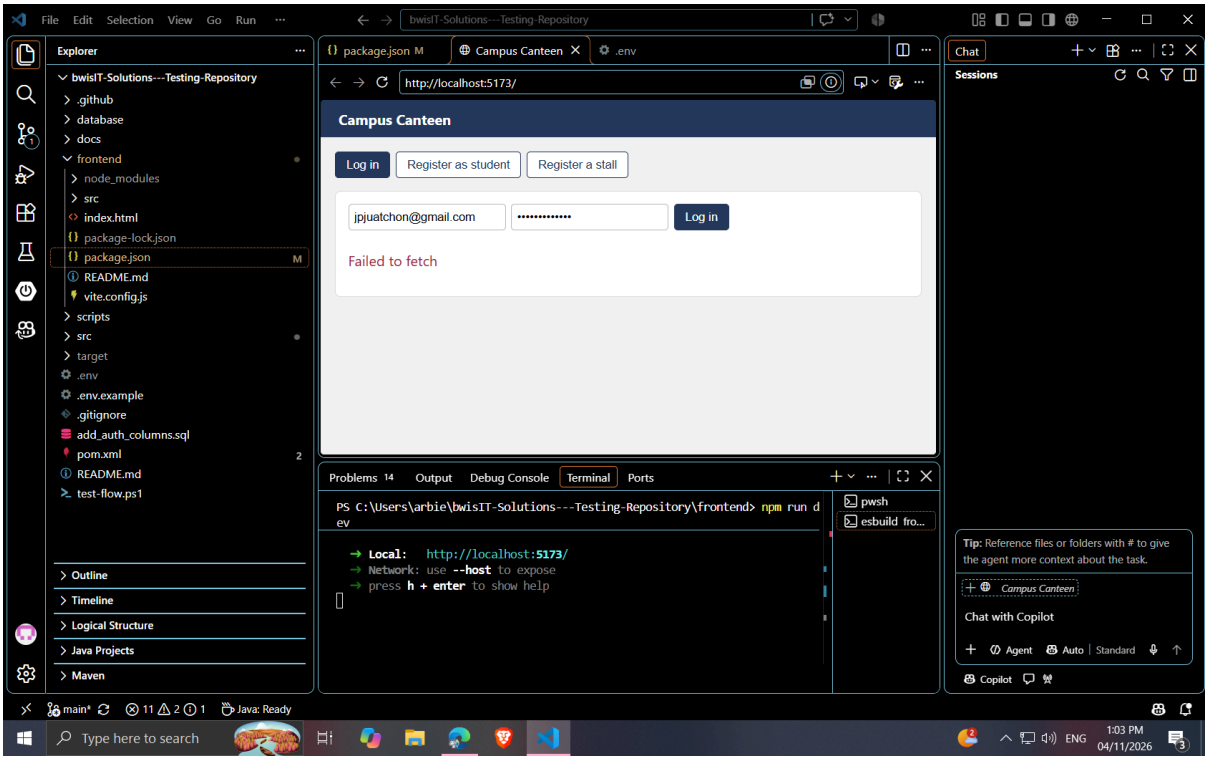
a. 

(Frontend login screen showcase)

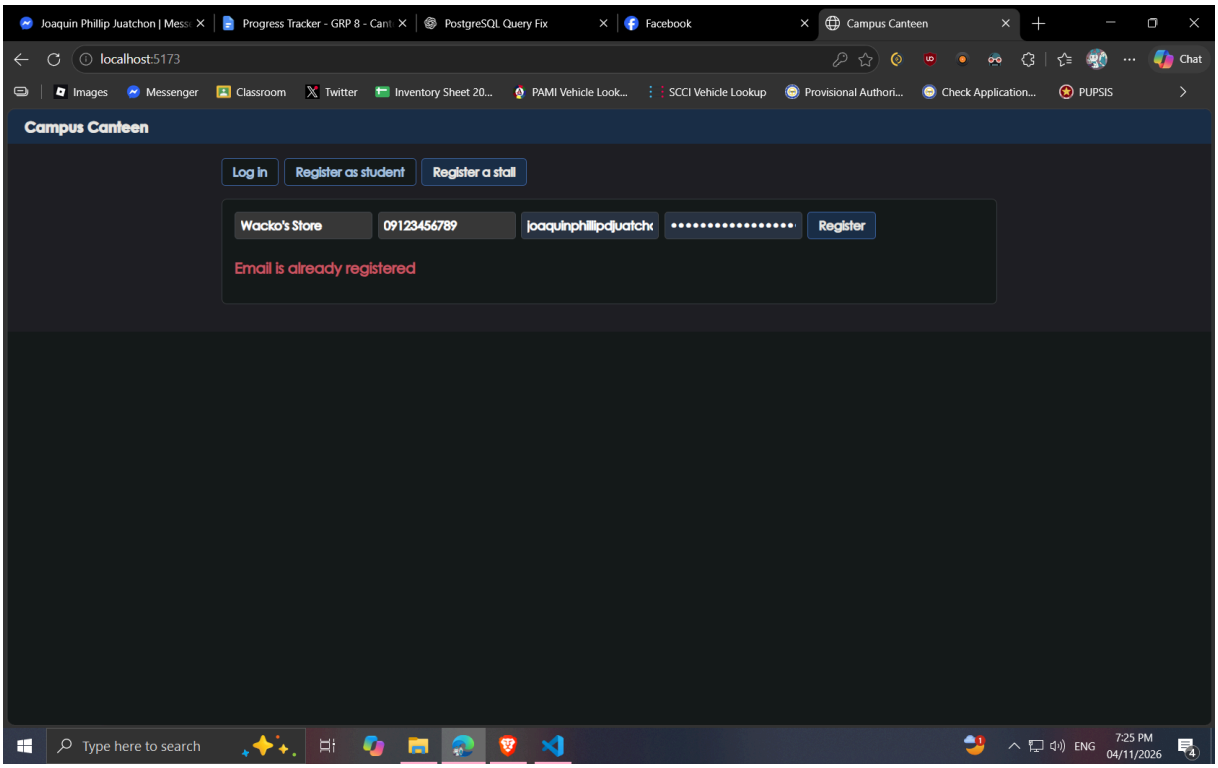
b. 

(Frontend screen showing the Student Home page)

4. November 4

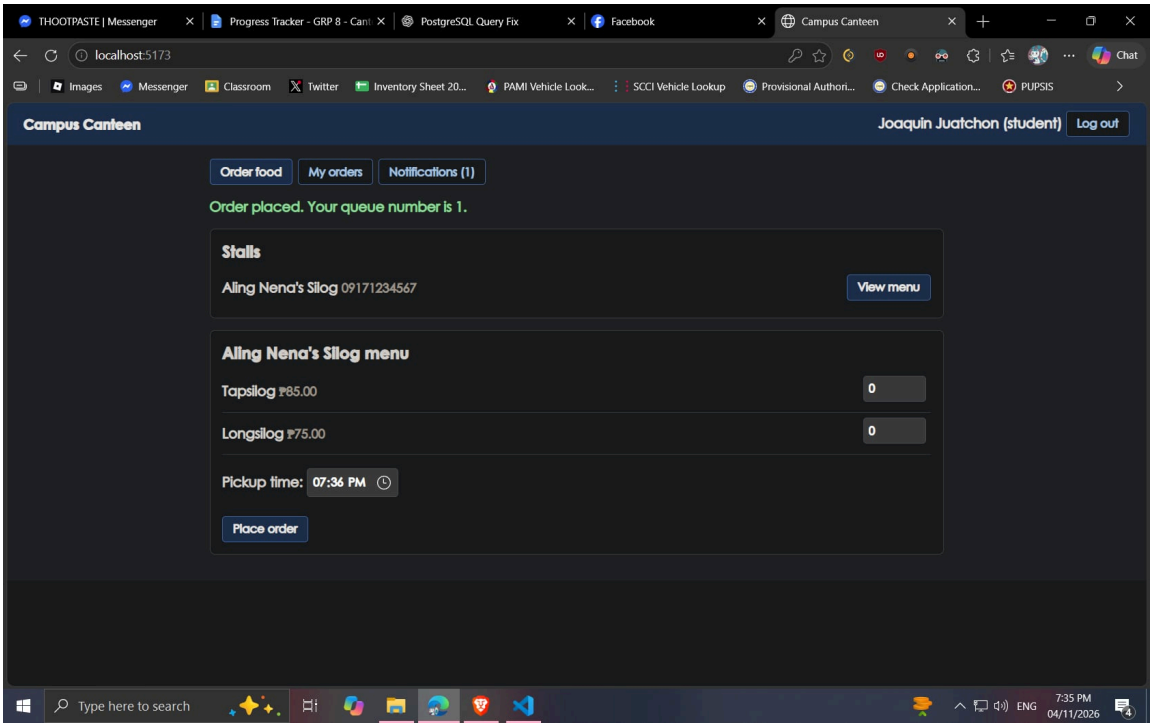
a. 

(Groupmate Juatchon showcasing application working on his computer)

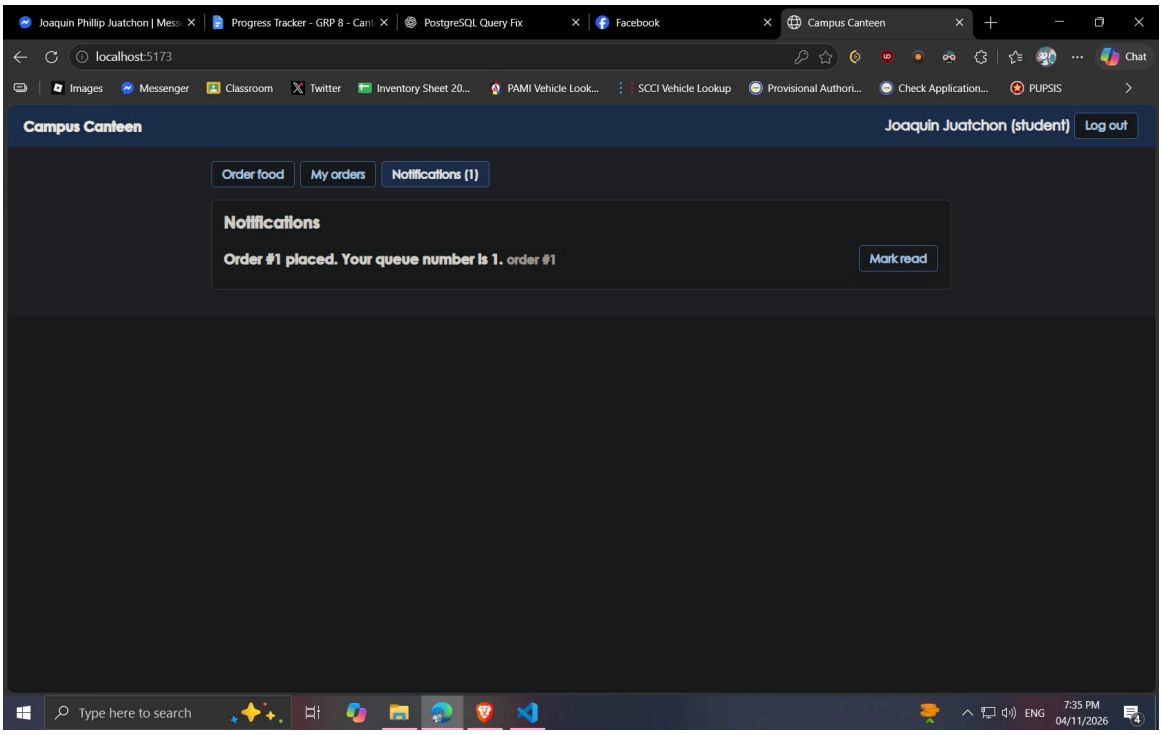
b. 

(Groupmate Juatchon attempting to register a stall using an existing email

that is already registered as a student)

c. 

(Groupmate Juatchon places an order and waits for the pickup time.)

d. 

(Notifications for the placed order.)

e.

```
postgres=# \du
                                   List of roles
      Role name       |                         Attributes
----------------------+------------------------------------------------------------
 canteen_admin        | Create role
 canteen_app          |
 canteen_app_grp      | Cannot login
 canteen_readonly     |
 canteen_readonly_grp | Cannot login
 postgres             | Superuser, Create role, Create DB, Replication, Bypass RLS


postgres=# SELECT
postgres-#     member.rolname AS member_role,
postgres-#     granted.rolname AS granted_group
postgres-# FROM pg_auth_members
postgres-# JOIN pg_roles AS member
postgres-#     ON pg_auth_members.member = member.oid
postgres-# JOIN pg_roles AS granted
postgres-#     ON pg_auth_members.roleid = granted.oid
postgres-# WHERE member.rolname IN (
postgres(#     'canteen_admin',
postgres(#     'canteen_app',
postgres(#     'canteen_readonly'
postgres(# )
postgres-# ORDER BY member_role, granted_group;
   member_role    |    granted_group
------------------+---------------------
 canteen_app      | canteen_app_grp
 canteen_readonly | canteen_readonly_grp
(2 rows)
```

(PostgreSQL roles and memberships (`\du` output), without exposing credentials.)

f.

```
canteen_db=# SELECT
canteen_db-#     tablename,
canteen_db-#     has_table_privilege('canteen_app', 'public.' || tablename, 'SELECT') AS app_select,
canteen_db-#     has_table_privilege('canteen_app', 'public.' || tablename, 'INSERT') AS app_insert,
canteen_db-#     has_table_privilege('canteen_app', 'public.' || tablename, 'UPDATE') AS app_update,
canteen_db-#     has_table_privilege('canteen_app', 'public.' || tablename, 'DELETE') AS app_delete,
canteen_db-#     has_table_privilege('canteen_app', 'public.' || tablename, 'TRUNCATE') AS app_truncate,
canteen_db-#     has_table_privilege('canteen_readonly', 'public.' || tablename, 'SELECT') AS readonly_select,
canteen_db-#     has_table_privilege('canteen_readonly', 'public.' || tablename, 'INSERT') AS readonly_insert,
canteen_db-#     has_table_privilege('canteen_readonly', 'public.' || tablename, 'UPDATE') AS readonly_update,
canteen_db-#     has_table_privilege('canteen_readonly', 'public.' || tablename, 'DELETE') AS readonly_delete,
canteen_db-#     has_table_privilege('canteen_readonly', 'public.' || tablename, 'TRUNCATE') AS readonly_truncate
canteen_db-# FROM pg_tables
canteen_db-# WHERE schemaname = 'public'
canteen_db-# ORDER BY tablename;
   tablename    | app_select | app_insert | app_update | app_delete | app_truncate | readonly_select | readonly_insert | readonly_update | readonly_delete | readonly_truncate
----------------+------------+------------+------------+------------+--------------+-----------------+-----------------+-----------------+-----------------+-------------------
 customer_order | t          | t          | t          | f          | f            | f               | f               | f               | f               | f
 menu_item      | t          | t          | t          | t          | f            | f               | f               | f               | f               | f
 notification   | t          | t          | t          | f          | f            | f               | f               | f               | f               | f
 order_item     | t          | t          | f          | f          | f            | f               | f               | f               | f               | f
 queue_entry    | t          | t          | t          | f          | f            | f               | f               | f               | f               | f
 student        | t          | t          | f          | f          | f            | f               | f               | f               | f               | f
 vendor         | t          | t          | t          | f          | f            | f               | f               | f               | f               | f
(7 rows)
```

(Privilege verification output showing role flags, table ownership/access, sequence access, and database connection privileges.)

g.

```
C:\Users\Marcos\Documents\programming stuff\datamana\bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System\db-access-control\tests\runtime-allowed.sql
PS C:\Users\Marcos\Documents\programming stuff\datamana\bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System> psql -U canteen_app -d canteen_db -f "C:\Users\Marcos\Documents\programming stuff\datamana\bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System\db-access-control\tests\runtime-allowed.sql"
Password for user canteen_app:

BEGIN
psql:C:/Users/Marcos/Documents/programming stuff/datamana/bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System/db-access-control/tests/runtime-allowed.sql:31: NOTICE:  PASS: runtime can read
 menu records
psql:C:/Users/Marcos/Documents/programming stuff/datamana/bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System/db-access-control/tests/runtime-allowed.sql:31: NOTICE:  PASS: runtime inserted
 an order
psql:C:/Users/Marcos/Documents/programming stuff/datamana/bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System/db-access-control/tests/runtime-allowed.sql:31: NOTICE:  PASS: runtime updated
its test order
DO
ROLLBACK
PS C:\Users\Marcos\Documents\programming stuff\datamana\bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System>
```

(Runtime allowed tests showing the PASS notices and final ROLLBACK.)

h.

```
C:\Users\Marcos\Documents\programming stuff\datamana\bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System\db-access-control\tests\runtime-denied.sql
PS C:\Users\Marcos\Documents\programming stuff\datamana\bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System> psql -U canteen_app -d canteen_db -f "C:\Users\Marcos\Documents\programming stuff\datamana\bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System\db-access-control\tests\runtime-denied.sql"
Password for user canteen_app:

psql:C:/Users/Marcos/Documents/programming stuff/datamana/bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System/db-access-control/tests/runtime-denied.sql:24: NOTICE:  PASS: runtime DELETE on
 student is denied
psql:C:/Users/Marcos/Documents/programming stuff/datamana/bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System/db-access-control/tests/runtime-denied.sql:24: NOTICE:  PASS: runtime access to
 protected pg_authid is denied
psql:C:/Users/Marcos/Documents/programming stuff/datamana/bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System/db-access-control/tests/runtime-denied.sql:24: NOTICE:  PASS: runtime lacks CON
NECT on unrelated postgres database
DO
PS C:\Users\Marcos\Documents\programming stuff\datamana\bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System>
```

(Runtime denied tests showing the PASS notices.)

i.

```
PS C:\Users\Marcos\Documents\programming stuff\datamana\bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System> psql -U canteen_readonly -d canteen_db -f ".\db-access-control\tests\readonly-allowed.sql"
Password for user canteen_readonly:

 item_id | vendor_id |   item_name   | price | is_available
---------+-----------+---------------+-------+--------------
       1 |         2 | Double Burger | 95.00 | t
       2 |         4 | Plain Waffle  | 50.00 | t
       3 |         2 | Cheeseburger  | 65.00 | t
       4 |         1 | Tapsilog      | 85.00 | t
       5 |         1 | Tosilog       | 80.00 | t
(5 rows)


 order_id | vendor_id |         order_date         |  status   | item_id | quantity | price_at_order
----------+-----------+----------------------------+-----------+---------+----------+----------------
        4 |         1 | 2026-11-04 21:06:56.935498 | cancelled |       4 |        1 |          85.00
        1 |         1 | 2026-11-04 20:29:54.654588 | pending   |       4 |        2 |          85.00
        2 |         1 | 2026-11-04 20:29:54.654588 | preparing |       8 |        1 |          75.00
(3 rows)


PS C:\Users\Marcos\Documents\programming stuff\datamana\bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System> psql -U canteen_readonly -d canteen_db -f ".\db-access-control\tests\readonly-denied.sql"
Password for user canteen_readonly:

psql:db-access-control/tests/readonly-denied.sql:40: NOTICE:  PASS: read-only INSERT is denied
psql:db-access-control/tests/readonly-denied.sql:40: NOTICE:  PASS: read-only UPDATE is denied
psql:db-access-control/tests/readonly-denied.sql:40: NOTICE:  PASS: read-only DELETE is denied
psql:db-access-control/tests/readonly-denied.sql:40: NOTICE:  PASS: read-only TRUNCATE is denied
psql:db-access-control/tests/readonly-denied.sql:40: NOTICE:  PASS: read-only access to unapproved student table is denied
DO
```

(Read-only allowed query results and denied-operation PASS notices.)

j.

```
PS C:\Users\Marcos\Documents\programming stuff\datamana\bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System> psql -U canteen_app -d canteen_db -c "CREATE DATABASE test_restricted_db;"
Password for user canteen_app:

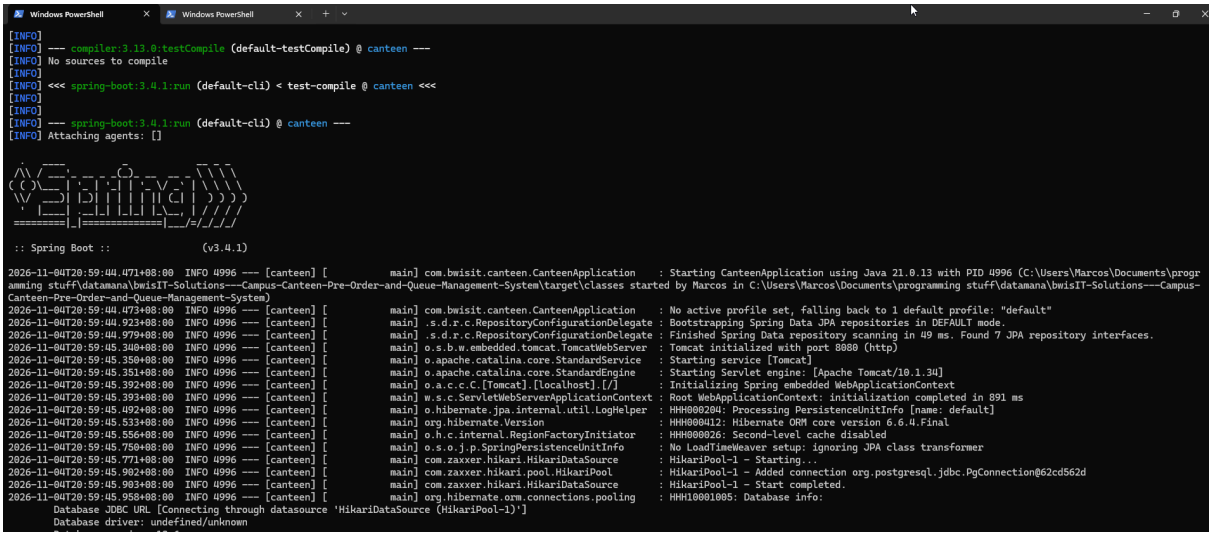
ERROR:  permission denied to create database
```

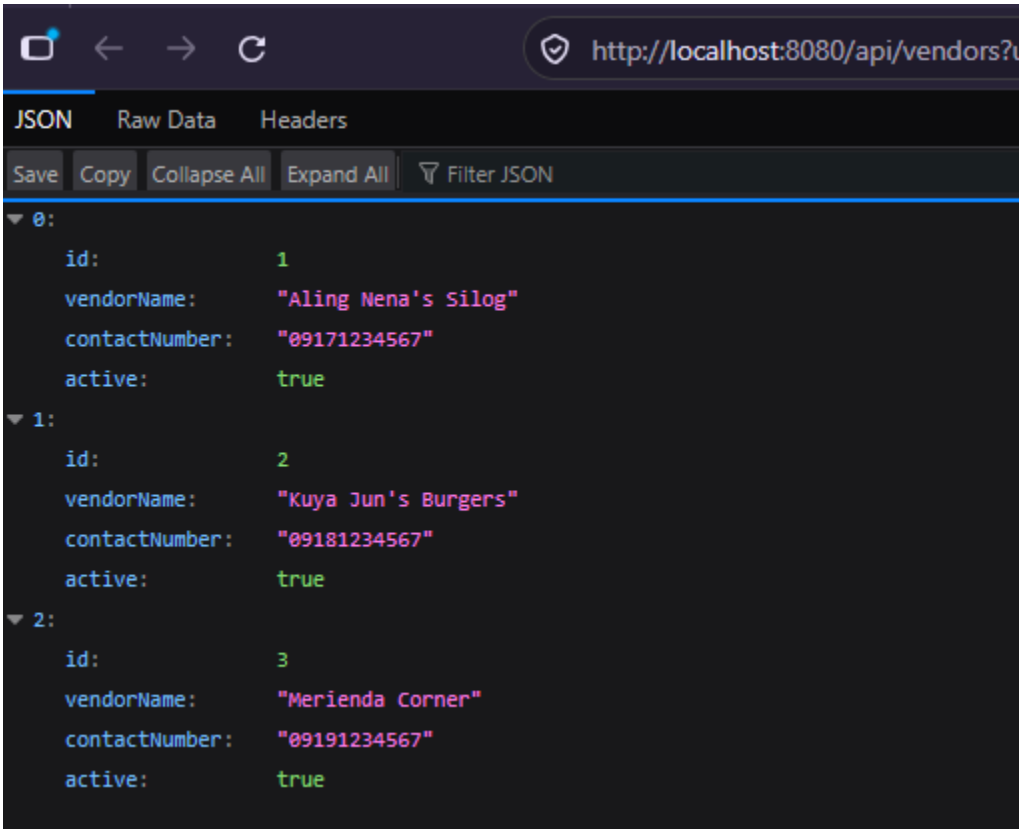
```
PS C:\Users\Marcos\Documents\programming stuff\datamana\bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System> psql -U canteen_app -d postgres
Password for user canteen_app:

psql: error: connection to server at "localhost" (::1), port 5432 failed: FATAL:  permission denied for database "postgres"
DETAIL:  User does not have CONNECT privilege.
```

(Denied database creation and denied connection to `postgres` as `canteen_app`.)

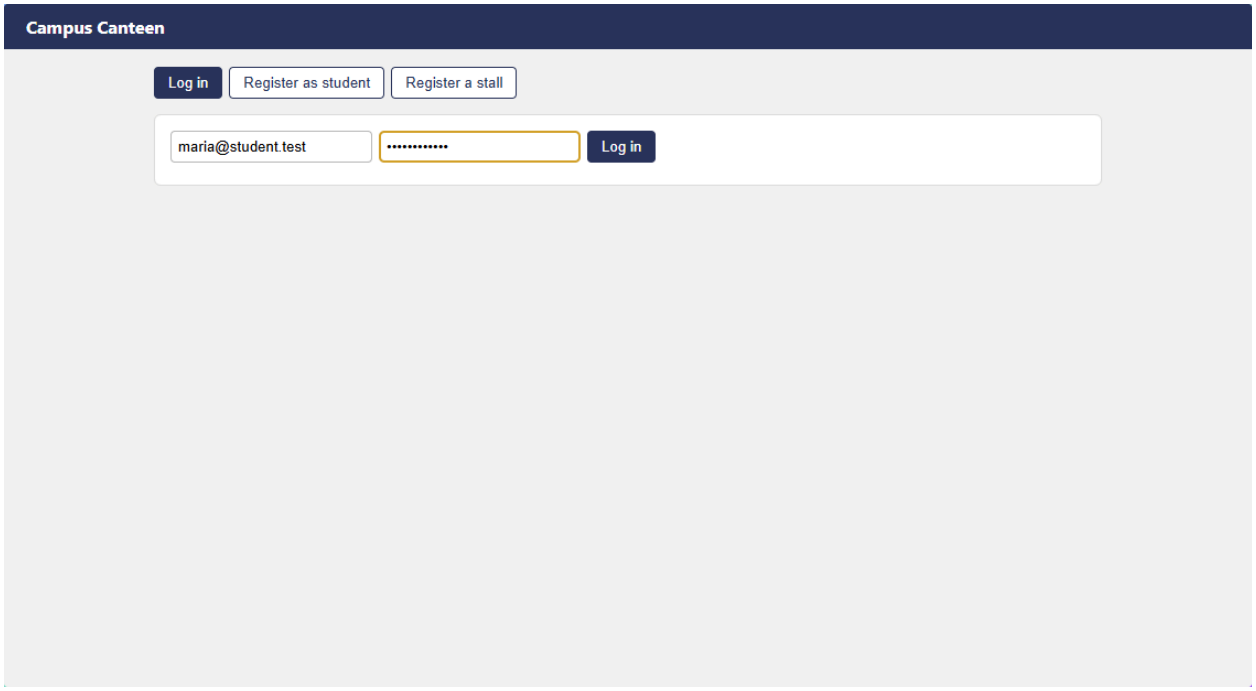
k.

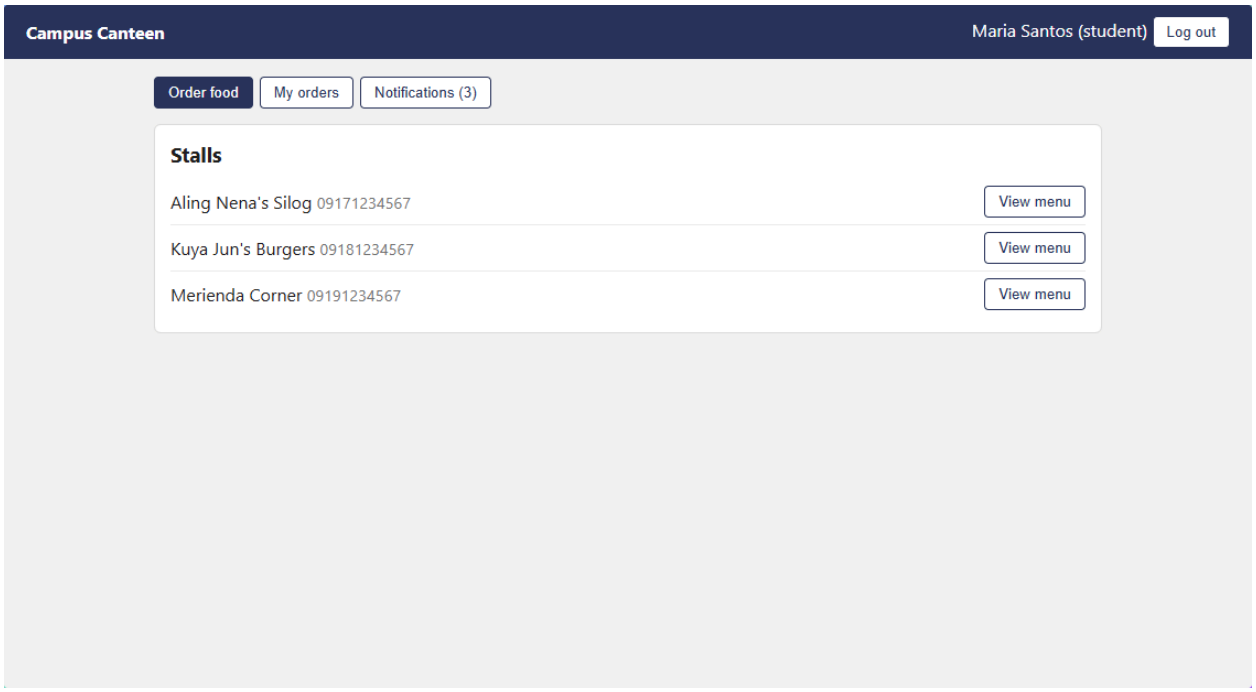


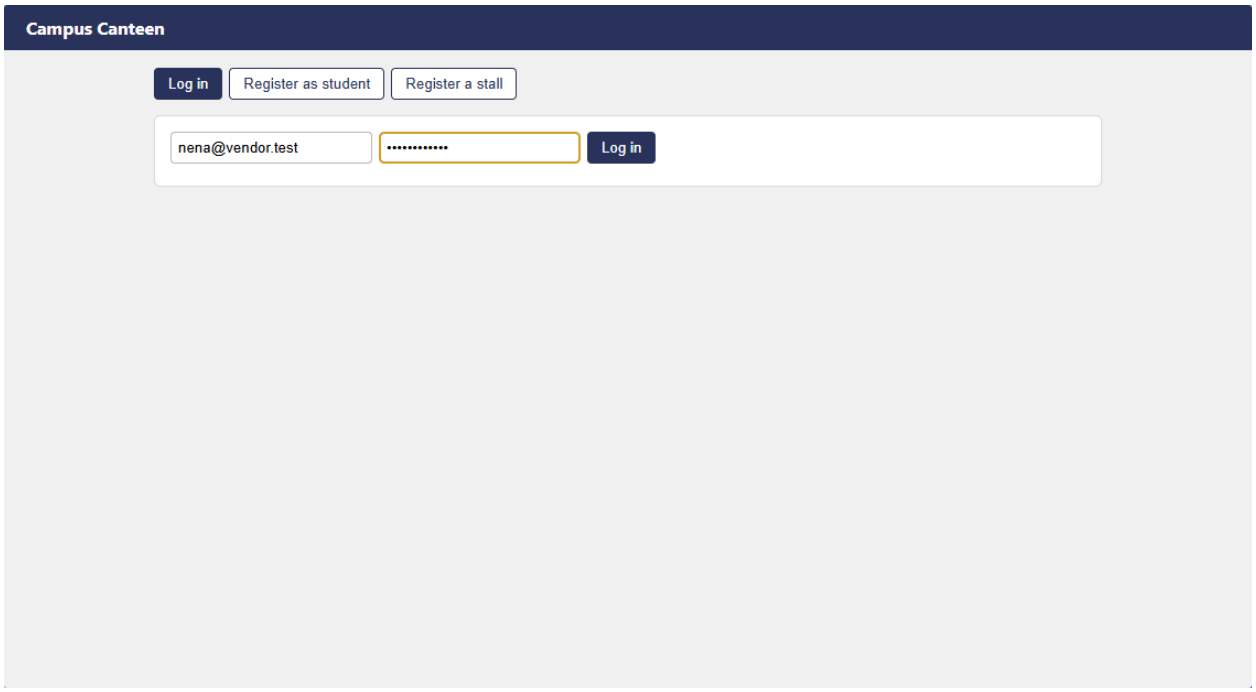


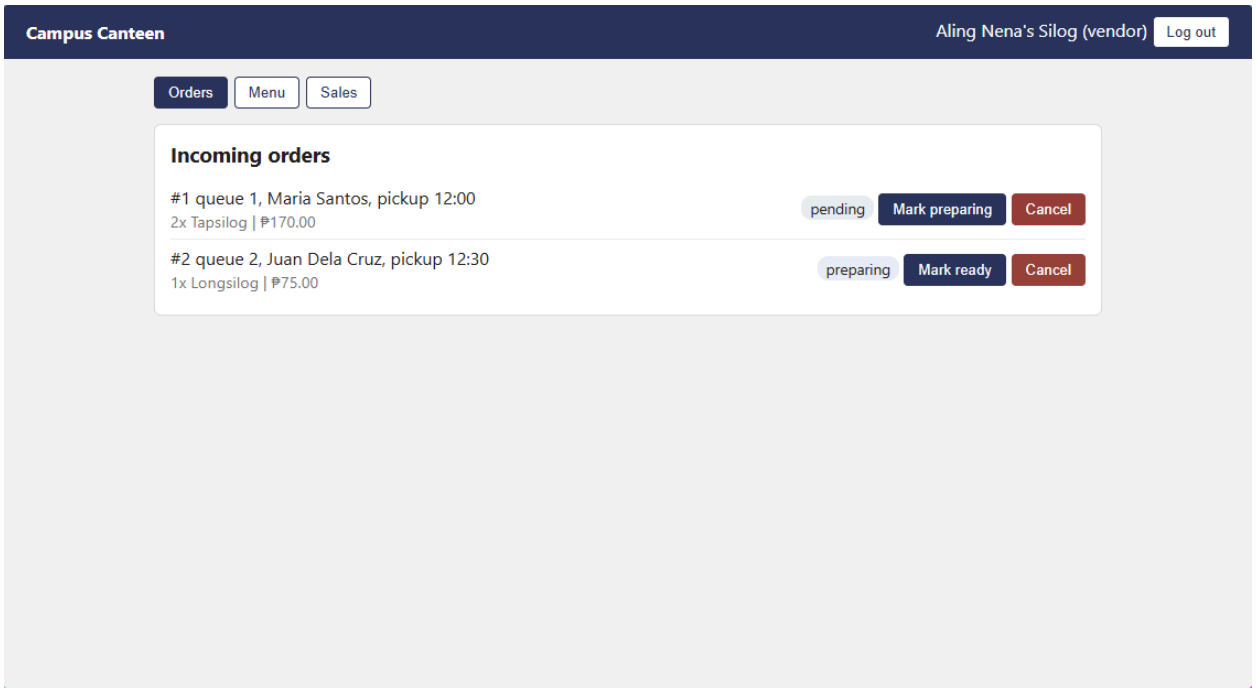
(Spring Boot Started message and separate `/api/vendors` request showing the three active stalls.)

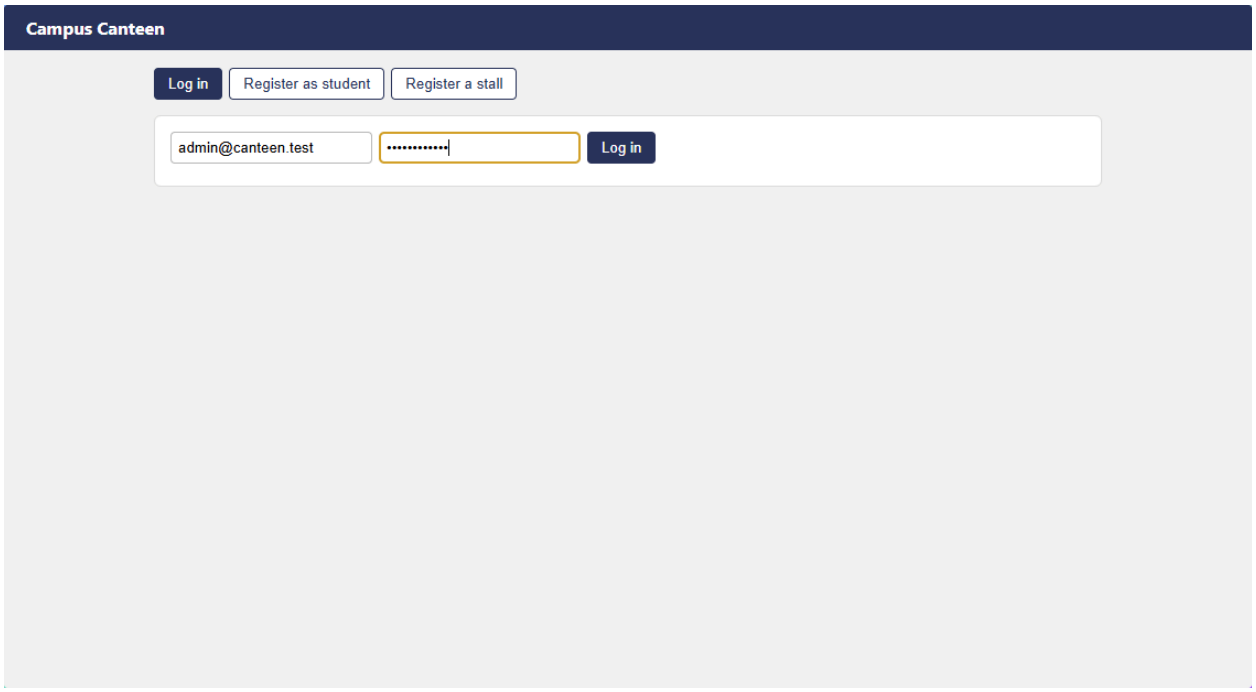
I.

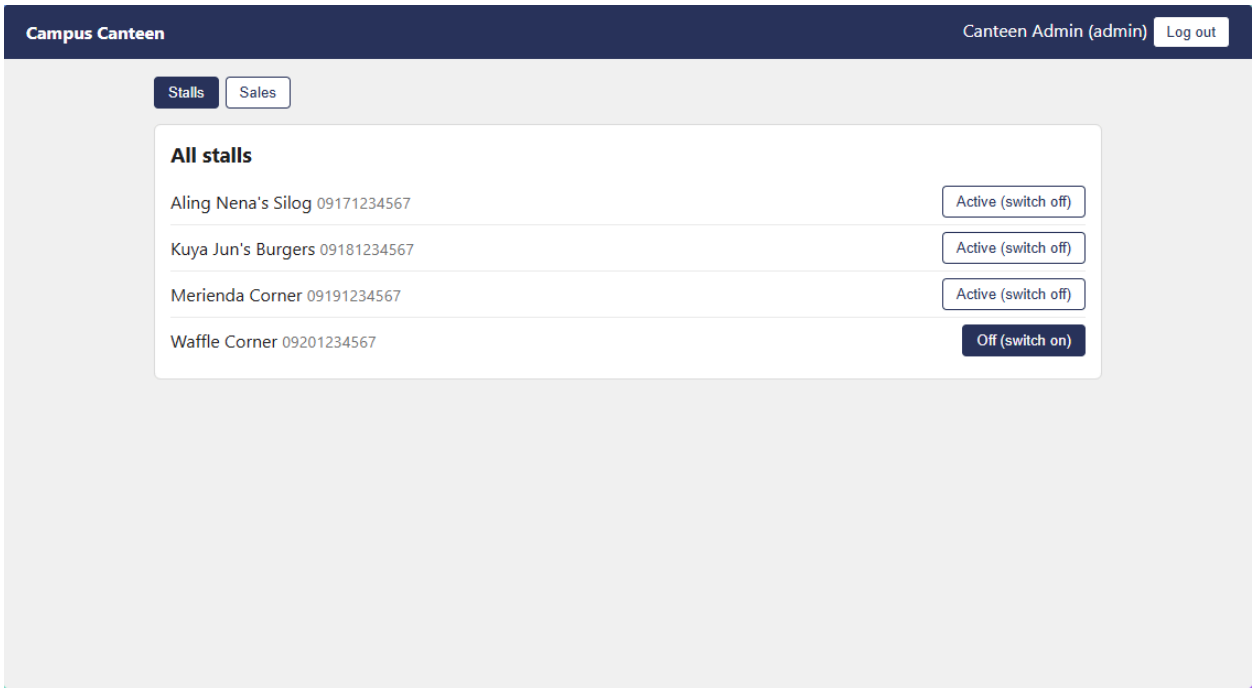

Campus Canteen
Log in
Register as student
Register a stall
maria@student.test
Log in


Campus Canteen
Maria Santos (student)
Log out
Order food
My orders
Notifications (3)
Stalls
Aling Nena's Silog 09171234567
View menu
Kuya Jun's Burgers 09181234567
View menu
Merienda Corner 09191234567
View menu


Campus Canteen
Log in
Register as student
Register a stall
nena@vendor.test
Log in


Campus Canteen
Aling Nena's Silog (vendor)
Log out
Orders
Menu
Sales
Incoming orders
#1 queue 1, Maria Santos, pickup 12:00
2x Tapsilog | ₱170.00
pending
Mark preparing
Cancel
#2 queue 2, Juan Dela Cruz, pickup 12:30
1x Longsilog | ₱75.00
preparing
Mark ready
Cancel





(Student menu/order and notification view; vendor orders/menu/sales view; administrator stalls/sales view. Browser evidence from `http://localhost:5173`, with no credentials or secrets visible)